

15 AI in the research process

The agents I used could understand a goal without pursuing it. They could correctly describe the result I wanted, yet direct their work toward a preliminary step or a different endpoint. This became a central difficulty as I delegated more substantial tasks. Their growing autonomy reduced the need for detailed mathematical and programming instructions, but left me with the problem of making their work serve the purpose for which I had requested it. This section develops my interpretation of that experience; the separate disclosure records the extent and provenance of AI use in the thesis.

15.1 An autonomous mathematical argument

The local maximality proof illustrates how much could be delegated. I asked a GPT-5-series model to develop a local maximality argument for HKO in facet coordinates. I suggested looking for several upper bounds and somehow showing that together they control a neighborhood after quotienting out the symmetries. I had neither specified a finite certificate nor worked out how to construct the upper bounds.

The model developed essentially the entire argument from that starting point. It identified a sufficient configuration of twenty-six upper bounds in the twenty-five transverse dimensions. An initial derivative-based approach had provided only twenty-five nonsingular cases. The model then used the weights β in the quadratic objective to construct feasible sections: an invertible 5×5 closure minor lets five weights depend smoothly on the facet parameters while the remaining weights stay fixed. This supplied upper bounds even where the initial approach failed. It also completed proof steps involving the volume and wrote the SageMath code for the exact computations. The resulting theorem controls arbitrary sufficiently small perturbations of the ten facets, including those that break the product structure.

15.2 Understanding a goal and pursuing it

Other tasks exposed a gap between what an agent could do and what it actually did. An agent could identify the work needed to finish a chapter and still return another plan for doing it. It could recognize that a calculation was only a preliminary step and nevertheless treat that step as the completion of its task. These failures sometimes looked oddly motivational. The problem was not always that the agent had inferred my goal incorrectly: it could state that goal and explain how its present work fell short. It still pursued something else.

I regard this as a manifestation of the alignment problem in ordinary research work. Understanding the user's objective and selecting an objective to pursue are distinct. Useful delegation does not require those objectives to be identical. It requires enough overlap that the work the agent actually pursues produces a useful end result for the user. A model might be trying to produce an answer that would receive a good evaluation; I might need a proof to settle a mathematical question or a chapter that a particular reader can follow. The practical issue is whether satisfying the former criterion also satisfies enough of the latter.

My hypothesis is that reinforcement learning with verifiable rewards encourages models to learn a concept of a *Grader*: something whose judgment of success their work is organized to satisfy. Such training uses outcomes that can be checked, for example a mathematical answer or a program's performance on test cases. DeepSeek-R1-Zero, for instance, was trained with rule-based accuracy and format rewards, without complete human demonstrations of the reasoning strategies it developed [GYZ+25]. This makes the Grader interpretation plausible to me, but does not establish it as an explanation of the agents I used. The interpretation concerns which criterion directs their actions, beyond whether they exploit a loophole in a test or obey an instruction too literally.

I therefore tried to make the desired result clear enough to evaluate: what had to be delivered, what it was needed for, and what would leave the task unfinished. The purpose was to bring the agent’s pursued criterion into sufficient overlap with my needs. Merely asking it to repeat those needs could not establish that this had happened. In choosing prompts, I also found it useful to distinguish two possible effects of the text: its wording may cue a learned pattern of behavior, while its content supplies information for deliberate reasoning. That distinction was a working interpretation, not a measured account of how a particular prompt changed the model’s behavior.

15.3 Evaluating the whole task

For a sustained task, the relevant checkpoint asks whether the current reasoning, evidence, and actions advance the right whole goal within the available budget. A component can be correct and useful while the requested result remains far from complete. This matters both during the work, when there is still time to change direction, and at its end, when a successful calculation or a plausible report can otherwise stand in for the result that was requested.

Time and computational resources belong in that evaluation. An investigation can produce useful information and still consume the time needed to finish the work. When the task is to deliver a readable chapter, identifying another possible improvement has to be weighed against writing and revising the chapter. In a mathematical search, the value of another computation depends on the question it can resolve and the resources left for following up. The criterion is the contribution to the intended result under these constraints.

This shifted much of my work from managing individual steps toward translating the needs of the project into objectives an agent could pursue and evaluate. I knew why I wanted a theorem, an experiment, or a thesis chapter; that purpose did not automatically accompany a request for one. Supplying it meant explaining the intended result, the reader or use it had to serve, and the resources available. The agent could then choose many of its own intermediate steps. My intervention was needed when those choices ceased to serve the task, even if each choice could be defended in isolation.

15.4 Why exposition remained difficult

The proof work in this MSc project was sufficiently tractable for the available models that I could often delegate it with little detailed direction. Producing passable mathematical exposition was much harder. This surprised me given the amount of written material used in language-model training. The later work on the thesis became a prolonged struggle with writing and task-management quirks, even when the models could partly recognize those quirks in discussion.

One recurring problem was prose that sounded polished while making its content difficult to recover. Inflated claims of significance, colorful phrasing, and abstract summaries gave passages an appearance of substance that their explanations did not support. This is what I mean by *AI slop* here. Another difficulty concerned the burden placed on the reader’s memory. With the relevant pages in its context, an agent could resolve chains of forward and backward references across several pages. A human reader had to remember what had been deferred, or interrupt the argument to recover an earlier definition. An agent’s ability to reconstruct the meaning from the full text therefore did not show that a reader could comfortably follow it in order.

I suspect that preference training contributes to the tendency toward impressive presentation. In reinforcement learning from human feedback, rankings of model outputs can be used to train toward preferred responses, as in InstructGPT [OWJ+22]. If an impressive presentation is easier to recognize in a brief comparison than a clear explanation is to assess through sustained reading, that preference signal may favor the appearance of quality. This is a hypothesis about a possible source of the prose I encountered, rather than a demonstrated causal account of those models’ training.

The practical difficulty persisted after I had named it. A model could acknowledge that a passage needed a concrete argument and then return another description of the argument’s intended role. It could recognize that the goal was a readable chapter while spending its effort on further diagnosis. I had to devise corrections tailored to recurring behaviors; there was no single instruction whose recognition reliably produced the desired writing. The ability to state the criticism did not settle what the agent would subsequently do.

References

- [GYZ+25] Daya Guo, Dejian Yang, Haowei Zhang, et al. “DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning”. In: *Nature* 645 (2025), pp. 633–638. DOI: 10.1038/s41586-025-09422-z.
- [OWJ+22] Long Ouyang, Jeff Wu, Xu Jiang, et al. *Training language models to follow instructions with human feedback*. Mar. 4, 2022. DOI: 10.48550/arXiv.2203.02155. arXiv: 2203.02155 [cs.CL]. URL: <https://arxiv.org/abs/2203.02155> (visited on 09/18/2026).